

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q.1 Solve Any Two of the following.

12

- A) A chartered accountant applies for a job in two firms X and Y. The probability of his being selected in the firm X is 0.7, and being rejected in the firm Y is 0.5, and the probability of at least one of his applications being rejected is 0.6. What is the probability that he will be selected in one of the firms?
- B) The factory F_1 produces 1000 articles, 20 of them being defective; the factory F_2 produces 4000 articles, 40 of them being defective and the factory F_3 produces 5000 articles, 50 of them being defective. If one article is chosen from all these articles put in one stockpile and is found to be defective, find the probability that it is from the factory F_1 .
- C) Two dice are rolled, find the probability of getting sum is equal to (i) 6 (ii) more or equal to 10

CO1

6

CO1

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CO1

6

Q.2 Solve Any Two of the following.

12

- A) A random variable X has the following distribution:

X:	1	2	3	4	5	6
P(X):	2k	4k	6k	8k	10k	12k

CO2

6

Determine (i) k (ii) $P(X < 4)$ (iii) $P(2 \leq X < 5)$ (iv) $E(X)$ (v) $\text{Var}(X)$

B) Fit the Binomial Distribution to the following data

X:	0	1	2	3	4
F:	28	62	46	10	4

CO2

6

C) In a sample of 1000 students, the mean and standard deviation of marks

Obtained by the students in a certain test are 14 and 2.5. Assuming the distribution to be normal find the number of students getting marks

CO2

6

(i) between 12 and 15, (ii) above 18, (iii) below 8

[Given : For a S.N.V. z area between $z = 0$ to $z = 0.4$ is 0.1554 , between $z = 0$ to $z = 0.8$ is 0.2881, that between $z = 0$ to $z = 1.6$ is 0.4452, between $z = 0$ to $z = 2.4$ is 0.4918]

Q.3 Solve Any Two of the following.

12

A) 10 students of B.Sc. Obtained following marks in statistics in internal assessment (X) and university examination (Y) as Calculate (i) the Karl Pearson's coefficient of correlation (ii) standard error (iii) probable error from the following data: <https://www.batuonline.com>

CO3

6

X:	6	8	12	15	18	20	24	28	31
Y:	10	12	15	15	18	25	22	26	28

B) Obtain Rank Correlation Coefficient (ρ), for the following data,

CO3

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X	68	64	75	50	64	80	75	40	55	64
Y	62	58	68	45	81	60	68	48	50	70

C) Test the significance of the correlation for the following value based on the number of observation :

CO3

6

i. 10 & ii. 100, $r = 0.4$ & $r = 0.9$

Q.4 Solve Any Two of the following.

12

A) The data about sales and advertisement expenditure of a firm is given below

6

	sales	advertisement expenditure
Mean	40	6
Standard Deviation	10	1.5

CO4

Coefficient of correlation, $r = 0.9$

a) Estimate the likely sales for a proposed advertisement expenditure of Rs 10 crores.

b) What should be the advertisement expenditure if the firm proposes a sales target of 60 crores of rupees?

B) Obtain the equation of the regression lines from the following data,

X	91	97	108	121	67	124	51	73	111	57
Y	71	75	69	97	70	91	39	61	80	47

CO4

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C) In a partially destroyed laboratory record, only the lines of regression of y on x and x on y are available as $4x - 5y + 33 = 0$ &

$20x - 9y = 107$ Respectively. Calculate $\bar{x}$ & $\bar{y}$ & the coefficient of correlation between x & y

CO4

6

Q. 5 Solve Any Two of the following.

12

A) A coin was tossed 484 times and the head turned up 265 times. Test the hypothesis that the coin is unbiased at 5% LOS.

CO5

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B) In a city A, 20% of a random sample of 900 school boys had a certain slight physical defect. In another city B, 18.5 of a random sample of 1600 school boys had the same defect. is the difference between the proportion significance?

CO5

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C) A sample of 100 electric bulbs produced by manufacturer A showed a mean life time of 1190 hours and a standard deviation of 90 hours. A sample of 75 bulbs produced by manufacturer B showed a mean life time of 1230 hours with the standard deviation of 120 hours. Is there a difference between the mean life time of two hands at significance level of i)5% and ii)1%

CO5

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*** End ***